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Studierichting: Informatica
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Bereken de volgende oneigenlijke integraal:

$$\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \frac{dx}{\sin^2(3x)}$$

onbepaalde integraal:

$$\int \frac{dx}{\sin^2(3x)}$$

Stel $u = 3x$, dan $du = 3\,dx$, dus $dx = \frac{1}{3}\,du$.

$$\int \frac{1}{\sin^2 u} \cdot \frac{1}{3} \cdot du$$

$$= \frac{1}{3} \int \csc^2 u \cdot du$$

$$= -\frac{1}{3} \cot u$$

$$= -\frac{1}{3} \cot(3x) + c$$

verticale asymptoot $x = \frac{\pi}{3}$ (omdat $\sin(3 \cdot \frac{\pi}{3}) = \sin(\pi) = 0$).

$$\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \frac{dx}{\sin^2(3x)} = \lim_{t \to \frac{\pi}{3}^-} \int_{\frac{\pi}{4}}^{t} \frac{dx}{\sin^2(3x)} + \lim_{s \to \frac{\pi}{3}^+} \int_{s}^{\frac{\pi}{2}} \frac{dx}{\sin^2(3x)}$$

eerste deel:

$$\lim_{t \to \frac{\pi}{3}^-} \left[-\frac{1}{3} \cot(3x) \right]_{\frac{\pi}{4}}^{t} = \lim_{t \to \frac{\pi}{3}^-} \left(-\frac{1}{3} \cot(3t) \right) - \left(-\frac{1}{3} \cot\left(\frac{3\pi}{4}\right) \right)$$

aangezien $3t \to \pi^-$, gaat $\cot(3t) \to -\infty$.

$$= \left(-\frac{1}{3} \cdot (-\infty) \right) - \left(-\frac{1}{3} \cdot (-1) \right) = +\infty - \frac{1}{3} = +\infty$$

tweede deel:

$$\lim_{s \to \frac{\pi}{3}^+} \left[-\frac{1}{3} \cot(3x) \right]_{s}^{\frac{\pi}{2}} = \left(-\frac{1}{3} \cot\left(\frac{3\pi}{2}\right) \right) - \lim_{s \to \frac{\pi}{3}^+} \left(-\frac{1}{3} \cot(3s) \right)$$

aangezien $3s \to \pi^+$, gaat $\cot(3s) \to +\infty$.

$$= 0 - \left(-\frac{1}{3} \cdot (+\infty) \right) = 0 - (-\infty) = +\infty$$

totaal:

$$\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \frac{dx}{\sin^2(3x)} = +\infty + (+\infty) = +\infty$$

References